



The Gender Conversion Gap

From STEM Participation to Labour-Market Power

A peer-review presentation for Estonia-facing gender-equality stakeholders. This project examines whether stronger women's STEM and digital participation indicators are associated with stronger labour-market and leadership outcomes across OECD countries — with Estonia as the focal case.

STATISTICS & DATA ANALYSIS CAPSTONE

38 OECD COUNTRIES

CROSS-COUNTRY ASSOCIATIONS

Research Design

Main Research Question

Do stronger STEM and digital participation indicators for women translate into better labour-market and leadership outcomes across OECD countries?

Estonia serves as the focal case throughout the analysis.

What the data can and cannot show:

- Compares **latest-available country-level indicators** across 38 OECD member countries
- Aggregate areas excluded from the sample
- Results are **cross-country associations**, not causal effects
- The "conversion gap" is a framing device — the data does not follow the same women from graduation into later career outcomes

Six Key Indicators

The analysis uses six OECD country-level indicators spanning education, digital participation, labour-market access, pay, work structure, and leadership.

1

Women among STEM graduates

Share of women among tertiary STEM graduates. Higher = stronger representation in STEM education.

2

Youth Coding Gap (M – F)

Male minus female gap in youth coding activity. Higher = larger male advantage.

3

Labour-Force Participation Gap (M – F)

Male minus female labour-force participation gap. Higher = larger male advantage.

4

Median gender wage gap

Median gender wage gap indicator. Higher = worse pay outcomes for women.

5

Part-Time Employment Gap (F – M)

Female minus male part-time employment incidence gap. Higher = women more concentrated in part-time work.

6

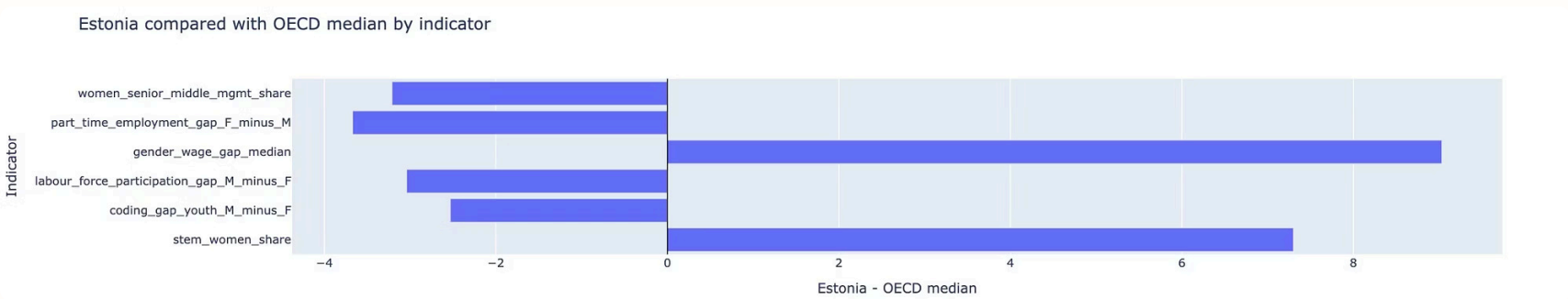
Women in Senior/Middle Management

Women's share in senior and middle management. Higher = stronger leadership representation.

Estonia vs OECD Median

Key descriptive finding: Estonia looks relatively strong in STEM education and some work-access indicators, but weaker in pay equality and senior/middle management representation.

Indicator	Estonia	OECD Median	Interpretation
Women among tertiary STEM graduates	41.55%	34.25%	Above OECD median
Youth coding gap, M – F	7.52 pp	10.05 pp	Smaller male advantage
Labour-force participation gap, M – F	4.92 pp	7.97 pp	Smaller gap than median
Median gender wage gap indicator	19.77%	10.74%	Substantially higher wage gap
Part-time employment gap, F – M	5.54 pp	9.21 pp	Smaller female part-time gap
Women in senior/middle management	31.51%	34.72%	Below OECD median



Four Hypotheses

H1 — STEM-to-Leadership Association

Expected: positive. Countries with higher shares of women among STEM graduates tend to have higher shares of women in senior/middle management.

H2 — STEM-to-Pay Equality

Expected: negative. Countries with higher shares of women among STEM graduates tend to have lower median gender wage gap values.

H3 — Digital Skills & Labour Participation (exploratory)

Expected: positive. Countries with larger male advantages in youth coding tend to have larger gender gaps in labour-force participation. Note: youth coding covers ages 16–24; labour-force participation covers 15–74.

H4 — Work-Structure Bottleneck (exploratory)

Expected: positive. Countries where women are more concentrated in part-time employment relative to men tend to have larger median gender wage gap indicator values.

Statistical Test Results — Summary

Pearson correlation and simple OLS are not treated as independent confirmations. OLS expresses the same linear relationship as an effect-size slope with confidence interval and R^2 . Spearman serves as a rank-based robustness check.

Hypothesis	Status	Pearson Result	Key Note
H1 — STEM-to-leadership	Partial support	$r = 0.43, p = 0.019$	Sensitive to Japan; Spearman weaker
H2 — STEM-to-pay equality	Not supported	$r = -0.17, p = 0.295$	Weak negative, not significant
H3 — Digital gap & participation	Not supported	$r = -0.52, p = 0.03$	Negative, not positive as expected
H4 — Part-time bottleneck	Not supported	$r = -0.21, p = 0.210$	Weak negative, not significant

❏ H2 and H4 were not supported. H3 showed a statistically significant association, but in the opposite direction to expectation. H1 showed partial support, but the result is sensitive to one influential country.

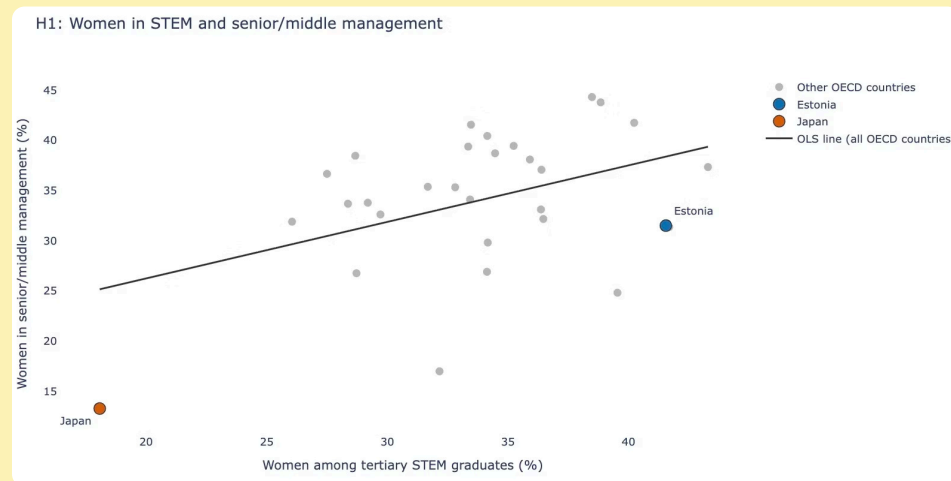
H1 — STEM-to-Leadership: Partial Support

Results (n = 30 OECD countries)

- Pearson $r = 0.43$, $p = 0.019$
- Spearman $\rho = 0.25$
- OLS slope = 0.56, 95% CI [0.10, 1.03]

There is some evidence of a positive linear association, but it is not robust enough to claim a stable general OECD-wide pattern.

⚠ Japan is highly influential. Japan has both very low women's STEM graduate share and very low women's senior/middle management share. When Japan is excluded, Pearson r drops to 0.17 and is no longer statistically significant.

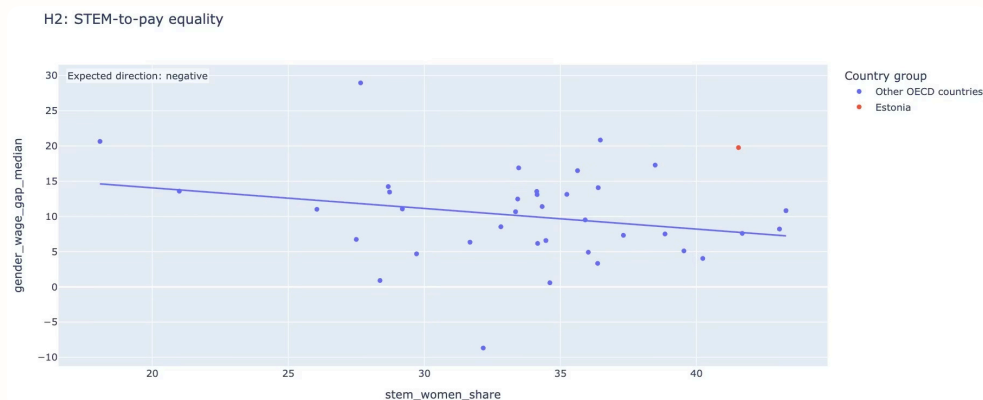


H2 & H4 — Not Supported

H2 — STEM-to-Pay Equality

n = 38 OECD countries. Pearson $r = -0.17$, $p = 0.295$. OLS slope = -0.21 , 95% CI $[-0.61, 0.19]$.

Key message: Higher women's STEM graduate share is not clearly associated with lower median gender wage gap values across OECD countries. This challenges a simple pipeline assumption — STEM education representation alone does not appear to be a reliable country-level marker of pay equality.



H4 — Work-Structure Bottleneck

n = 36 OECD countries. Pearson $r = -0.21$, $p = 0.210$. OLS slope = -0.14 , 95% CI $[-0.36, 0.08]$.

Key message: The simple country-level relationship between women's relative concentration in part-time employment and the median gender wage gap is weak and not statistically significant.

This does not prove that part-time work is irrelevant to pay inequality. The dataset does not measure occupational segregation, wage-setting institutions, parental leave systems, seniority patterns, or within-sector career progression.

H3 — Post-Mortem: Opposite Direction

Results (n = 32 OECD countries)

- Pearson $r = -0.52$, $p = 0.003$ (expected positive)
- Spearman $\rho = -0.46$, $p = 0.008$
- After excluding year-gap >2 years: $r = -0.48$, $p = 0.010$
- Exact-year subset (n = 25): $r = -0.36$, $p = 0.075$

The expected positive association is not supported. The negative direction remains in sensitivity checks, but becomes weaker in the exact-year subset.

Interpretation

Youth coding gender gaps and overall labour-force participation gaps may reflect different country-level mechanisms.

Coding is a specific youth digital activity indicator (ages 16–24). Labour-force participation is a broad working-age labour-market access indicator (ages 15–74).

Digital participation gaps should not be treated as the same mechanism as general labour-market participation gaps.

From Pipeline Metrics to Outcome Accountability

Main Insight

The analysis does not support a simple "more women in STEM = more labour-market equality" story. **STEM participation is not enough as a success metric.** The key issue is whether education and participation translate into pay equality, retention, progression, and leadership.

For Estonia: strong STEM graduate share and low participation gap, but high wage gap and below-median management representation — suggesting a possible downstream mismatch after education.

Recommendations for Estonia-Facing Actors

- **Measure the conversion layer** — Track women from STEM education into employment, pay outcomes, and leadership.
- **Focus on early-career progression** — Support mentoring, retention, and promotion pathways after graduation.
- **Encourage employer-level audits** — Review pay gaps, promotion rates, and representation in technical and management tracks.
- **Separate education from outcome metrics** — Do not use STEM graduate share alone as a success indicator.
- **Use Estonia as a focal monitoring case** — Strong STEM representation but weaker pay and leadership outcomes makes it a useful study case.

Limitations

- Cross-country associations, not causal effects
- Latest-available indicators are not all from the same year
- Median gender wage gap is not adjusted for occupation, hours, seniority, or sector

Possible Causes to Investigate

The current dataset does not directly test workplace mechanisms, but the mismatch points to possible downstream causes:

- hiring and first-job transition after STEM education;
- retention in STEM-intensive sectors;
- promotion into senior or management roles;
- pay-setting and pay transparency practices;
- cultural expectations around care, leadership, and technical roles.